BIOS Bioprocess model

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1 System and model description

This model aims to describe a growth system of *Pseudomonas putida* in different set-ups. The model not only studies the microorganism's growth dynamics but also the consumption of a carbon source (substrate) and other nutrients (chemical elements). Additionally, it examines the dynamics of gases that may be used in the process, including both those in the gas phase of the system and those dissolved in the liquid phase.

The model currently includes 22 equations, which are described in Section 2. To study growth, we use a basic Monod model, in which we assume that the microorganism's growth depends solely on the substrate:

$$\mu = \mu_{max} \cdot \frac{Subs}{K_{subs} + Subs} \tag{1}$$

For the study of nutrient dynamics, the equations follow the structure below:

$$\frac{dNutrient}{dt} = Feed - Sampling + Consumption/Generation \tag{2}$$

The different terms are described below:

1. **Feed**. This is the amount of nutrients introduced into the system through the feed. If the system is being fed, this term consists of the concentration of the nutrient in the feeding medium (mol/L) multiplied by the flow rate (L/min).

2. **Sampling**. This defines the decrease in the amount of nutrients due to the sampling events performed on the system. This term is defined as the concentration of the nutrient in the medium, i.e., the amount of nutrient divided by the volume of medium in the system, and is multiplied by the sampling flow rate.

3. **Consumption/Generation**. This term defines how the organism consumes or produces nutrients. It consists of the inverse of the yield factor (mol of nutrient consumed or generated per gram of biomass), multiplied by the growth rate and the biomass.

Regarding the equations related to gases, we must distinguish between working in the liquid phase or the gas phase, as there are differences. The gas-phase equations follow the structure below:

$$\frac{dXgas}{dt} = InGasFeed - Transfer - OffGas \tag{3}$$

The terms are as follows:

1. **InGasFeed**. This is the amount of gas entering the system through the air flow supplied. It consists of the concentration of that gas in the incoming air (mol/L) multiplied by the rate at which the air enters the system (L/min).
2. **Transfer**. This term describes how the gases in the air transfer into the culture medium. It consists of two parts: the KLa parameter, which describes the rate at which gases transfer from the gas phase to the liquid phase (a parameter to be estimated), and the concentration difference of the gas (gas saturation concentration minus the concentration of the gas in the liquid).
3. **Offgas**. This term describes the gas that leaves the system through the outgoing air flow. It is calculated as the concentration of the gas in the reactor's gas phase (mol/L) multiplied by the rate at which air exits the system (L/min).

La ecuación que describe el comportamiento de los gases en la fase líquida es la siguiente:

$$\frac{dXliquid}{dt} = Transfer + Consumption/Generation \tag{4}$$

The terms are as follows:

1. **Transfer**. This is exactly the same term used in the gas-phase equations.
2. **Consumption/Generation**. This term is exactly the same as the one used in the nutrient equations.

It is worth noting that in the case of CO_2, an additional term would be included, since in the aqueous phase this gas is hydrated to form H_2CO_3. However, this is not yet included in the model and will be incorporated later.

2 Model equations

This section presents the system equations. These equations may vary as the model continues to develop.

$$\frac{dSubs}{dt} = CSubs_f \cdot V_f - \left(\frac{Subs}{V_r}\right) \cdot V_s - \frac{1}{Y_{subs}} \cdot \mu \cdot Biomass \tag{5}$$

$$\frac{dBiomass}{dt} = \mu \cdot Biomass - \left(\frac{Biomass}{V_l}\right) \cdot V_s \tag{6}$$

$$\frac{dCa}{dt} = CCa_f \cdot V_f - \left(\frac{Ca}{V_l}\right) \cdot V_s - \frac{1}{Y_{Ca}} \cdot \mu \cdot Biomass \tag{7}$$

$$\frac{dCl}{dt} = CCl_f \cdot V_f - \left(\frac{Cl}{V_l}\right) \cdot V_s - \frac{1}{Y_{Cl}} \cdot \mu \cdot Biomass \tag{8}$$

$$\frac{dCo}{dt} = CCo_f \cdot V_f - \left(\frac{Co}{V_l}\right) \cdot V_s - \frac{1}{Y_{Co}} \cdot \mu \cdot Biomass \tag{9}$$

$$\frac{dCu}{dt} = CCu_f \cdot V_f - \left(\frac{Cu}{V_l}\right) \cdot V_s - \frac{1}{Y_{Cu}} \cdot \mu \cdot Biomass \tag{10}$$

$$\frac{dFe}{dt} = CFe_f \cdot V_f - \left(\frac{Fe}{V_l}\right) \cdot V_s - \frac{1}{Y_{Fe}} \cdot \mu \cdot Biomass \tag{11}$$

$$\frac{dMg}{dt} = CMg_f \cdot V_f - \left(\frac{Mg}{V_l}\right) \cdot V_s - \frac{1}{Y_{Mg}} \cdot \mu \cdot Biomass \tag{12}$$

$$\frac{dMo}{dt} = CMo_f \cdot V_f - \left(\frac{Mo}{V_l}\right) \cdot V_s - \frac{1}{Y_{Mo}} \cdot \mu \cdot Biomass \tag{13}$$

$$\frac{dNa}{dt} = CNa_f \cdot V_f + CNa_b \cdot Vb - \left(\frac{Na}{V_l}\right) \cdot V_s - \frac{1}{Y_{Na}} \cdot \mu \cdot Biomass \tag{14}$$

$$\frac{dZn}{dt} = CZn_f \cdot V_f - \left(\frac{Zn}{V_l}\right) \cdot V_s - \frac{1}{Y_{Zn}} \cdot \mu \cdot Biomass \tag{15}$$

$$\frac{dK}{dt} = CK_f \cdot V_f - \left(\frac{K}{V_l}\right) \cdot V_s - \frac{1}{Y_K} \cdot \mu \cdot Biomass \tag{16}$$

$$\frac{dNi}{dt} = CNi_f \cdot V_f - \left(\frac{Ni}{V_l}\right) \cdot V_s - \frac{1}{Y_{Ni}} \cdot \mu \cdot Biomass \tag{17}$$

$$\frac{dNH4}{dt} = CNH_{4f} \cdot V_f - \left(\frac{NH_4}{V_l}\right) \cdot V_s - \frac{1}{Y_{NH_4}} \cdot \mu \cdot Biomass \tag{18}$$

$$\frac{dP}{dt} = CP_f \cdot V_f - \left(\frac{P}{V_l}\right) \cdot V_s - \frac{1}{Y_P} \cdot \mu \cdot Biomass \tag{19}$$

$$\frac{dS}{dt} = CS_f \cdot V_f - \left(\frac{S}{V_l}\right) \cdot V_s - Y_S \cdot \mu \cdot Biomass \tag{20}$$

$$\frac{dCO2_l}{dt} = [K_L a \cdot (CO2_{sat} - CO2_l) \cdot V_l] + \frac{1}{Y_{CO2}} \cdot \mu \cdot Biomass \tag{21}$$

$$\frac{dCO2_g}{dt} = CO2_{Gasin} \cdot V_{Gas_{in}} - [K_L a \cdot (CO2_{sat} - CO2_l) \cdot V_l] - \left(\frac{CO2_g}{V_g}\right) \cdot V_{OffGas} \tag{22}$$

$$\frac{dO2_l}{dt} = K_L a \cdot (O2_{sat} - O2_l) \cdot V_l - \frac{1}{Y_{O2}} \cdot \mu \cdot Biomass \tag{23}$$

$$\frac{dO2_g}{dt} = O2_{Gasin} \cdot V_{Gas_{in}} - [K_L a \cdot (O2_{sat} - O2_l) \cdot V_l] - \left(\frac{O2_g}{V_g}\right) \cdot V_{OffGas} \tag{24}$$

$$\frac{dH}{dt} = CH_f \cdot V_f + CH_a \cdot Va - \left(\frac{H}{V_l}\right) \cdot V_s + \frac{1}{Y_H} \cdot \mu \cdot Biomass \tag{25}$$

$$\frac{dOH}{dt} = COH_f \cdot V_f + COH_b \cdot Vb - \left(\frac{OH}{V_l}\right) \cdot V_s \tag{26}$$

3 States os the system

The system states include biomass, the various nutrients in the medium broken down into their chemical elements, and the gaseous composition supplied to the culture (considering only oxygen and carbon dioxide). All system states are expressed in molar units, except for biomass, which is measured in grams of microorganism.

In this model, we use time units in minutes; therefore, the units of the differential equations are mol/min, except for biomass, which is expressed in g/min.

State	**Meaning**	**Units**
Subs	Substrate	mol
Biomass	Biomass	g
Ca	Calcium	mol
Cl	Chlorine	mol
Co	Cobaltium	mol
Cu	Cupper	mol
Fe	Iron	mol
Mg	Magnesium	mol
Mo	Molibdenum	mol
Na	Sodium	mol
Zn	Zinc	mol
K	Potassium	mol
Ni	Niquel	mol
NH_4	Ammonia	mol
P	Phosphorus	mol
S	Sulphate	mol
CO_2l	CO_2 in the liquid phase	mol
CO_2g	CO_2 in the gas phase	mol
O_2l	O_2 in the liquid phase	mol
O_2g	O_2 in the gas phase	mol
H	Protons	mol
OH	Hydroxils	mol

Table 1: States of the model/system

4 Secondary Terms and Equations of the Model

4.1 Terms Related to Volumes and Flow Rates

Term	Meaning	Units
V_l	Liquid volume (volume of the liquid phase)	L
V_g	Gas volume (volume of the gas phase)	L
V_f	Media feed flow rate	L/min
V_s	Sampling flow rate	L/min
V_{GasIn}	Gas feed flow rate	L/min
V_{OffGas}	Volumetric flow rate of gas leaving the system	L/min
V_a	Acid flow rate	L/min
V_b	Base flow rate	L/min

Table 2: Terms related to volumes and flow rates

4.2 Terms and secondary equations related to gases

4.2.1 Terms related to gases

Term	Meaning	Value/Calculation	Units
R	Rydberg's constant	8.31451	kJ/mol/K
T_{sc}	Temperature scale constant	273.15	K
$T_{Celsius}$	Temperature in Celsius	Experiment dependent	°C
T_{Kelvin}	Temperature in Kelvin	Calculated	K
Pr	Pressure	101325	Pa
$FractionO_2$	Fraction of O_2 in air	0.21	Dimensionless
$FractionCO_2$	Fraction of CO_2 in air	0.000407	Dimensionless
H_{CO_2}	Henry's constant for CO_2	3.4×10^{-7}	mol/L Pa
H_{O_2}	Henry's constant for O_2	1.3×10^{-8}	mol/L Pa
$PartialPressureO_2$	Partial pressure of O_2	Calculated	Pa
$PartialPressureCO_2$	Partial pressure of CO_2	Calculated	Pa
$CSatCO_2$	Saturation concentration of CO_2	Calculated	mol/L
$CSatO_2$	Saturation concentration of O_2	Calculated	mol/L
n	Number of gas moles O_2	Calculated	mol
V_{gas}	Gas volume (volume of gas phase)	Experiment dependent	L

Table 3: Terms related to gases

4.2.2 Equations related to gases

Ideal gas law:

$$Pr \cdot V_{gas} = n \cdot R \cdot T_{Kelvin} \tag{27}$$

From this equation, we obtain the number of moles of gas present:

$$n = \frac{Pr \cdot V_{gas}}{R \cdot T_{Kelvin}} \tag{28}$$

Now, we calculate the partial pressures of the gases:

$$PartialPressureO_2 = FractionO_2 \cdot Pr \tag{29}$$

$$PartialPressureCO_2 = FractionCO_2 \cdot Pr \tag{30}$$

Next, we calculate the saturation concentrations of the gases:

$$CSat_{O_2} = PartialPressureO_2 \cdot H_{O_2} \tag{31}$$

$$CSat_{CO_2} = PartialPressureCO_2 \cdot H_{CO_2} \tag{32}$$

It is important to note that the initial condition for the gases in the liquid phase will be equal to their saturation concentration; however, since we are working with moles and not moles per liter, we must multiply by the liquid phase volume (V_l):

$$IC^l_{CO_2} = CSat_{CO_2} \cdot V_l \tag{33}$$

$$IC^l_{O_2} = CSat_{O_2} \cdot V_l \tag{34}$$

On the other hand, if we want to know the initial condition for gases in the gas phase:

$$IC^g_{O_2} = FractionO_2 \cdot n \tag{35}$$

$$IC^g_{CO_2} = FractionCO_2 \cdot n \tag{36}$$

5 Parameters to estimate

These are the model parameters that must be estimated:

Parameter	Meaning	Unit
μ_{max}	Maximum growth rate	1/min
K_{subs}	Substrate uptake constant	g/L
KLa	Gas transfer coefficient	1/min

Table 4: Parameters to estimate